

GoalKeeper

Product Requirements Document & Technical Analysis

Turning Everyday Tasks into an Epic Co-op Adventure

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Abstract

GoalKeeper is a Progressive Web Application that transforms the mundane experience of managing personal goals and daily tasks into an immersive, cooperative gaming journey. By combining structured goal management with social accountability mechanics, gamification loops, and a rich reward economy, GoalKeeper aims to be the productivity companion people actually want to use. This document defines the full product vision, user stories, functional and non-functional requirements, system architecture, API surface, data models, and the phased delivery roadmap.

1. Executive Summary

GoalKeeper bridges the gap between productivity tooling and social gaming. Current goal-tracking applications are either too rigid and clinical (e.g., task managers) or too gamified without substance (habit streaks). GoalKeeper offers a hybrid: a seriously powerful goal and task management engine wrapped in a cooperative, community-driven gaming layer.

1.1 The Problem

- 68% of people abandon their personal goals within the first month due to lack of accountability and motivation (source: psychological research on habit formation).
- Existing productivity apps lack the social glue required to sustain long-term behavioural change.
- No current solution offers true cooperative accountability with a verifiable, fair XP and reward economy.

1.2 The Solution

- Long-horizon Goals broken into granular, scheduled Tasks — both one-time and recurring.
- Community-first accountability: buddy assignment, round-robin check-ins, and proof-of-completion verification.
- A fair XP economy where task completion and peer verification are both rewarded at configurable rates.
- A cosmetic reward store (titles, avatars, backgrounds) that makes progress feel tangible.
- A Solo mode for private goals, with reduced but meaningful rewards to encourage community participation.

1.3 Target Audience

Primary Users	Individuals aged 18-40 with personal improvement goals (fitness, learning, finance, wellness).
Secondary Users	Community managers, coaches, and team leads who want group accountability tools.
Tertiary Users	Enterprises and wellness programs seeking white-label or API-integrated accountability solutions.

1.4 Key Differentiators

- Cooperative XP system: buddies earn verifier points — alignment of incentives.
- Proof-of-completion: media-backed, human-verified task checking.
- Open Goal Library: community-sourced, curated goal and task templates.

- Calendar drag-and-drop scheduling integrated with task cards.
- AI-ready architecture for future intelligent insights and adaptive goal coaching.

2. Product Vision & Goals

2.1 Vision Statement

"Make personal achievement a shared adventure. GoalKeeper turns solo ambition into community-powered momentum."

2.2 Product Goals

G-01	Provide a best-in-class goal and task management UX on mobile-first PWA.
G-02	Deliver a community system with buddy accountability, leaderboards, and XP fairness.
G-03	Build a scalable backend (Golang) that supports millions of users and future AI integrations.
G-04	Launch a reward economy that drives engagement without pay-to-win mechanics.
G-05	Ship a configurable, API-first backend suitable for enterprise and third-party integrations.
G-06	Achieve an app store rating of 4.5+ within 6 months of launch.

2.3 Success Metrics

Metric	Description	Target (6mo)	Target (12mo)
MAU	Monthly Active Users	25,000	150,000
DAU/MAU	Daily engagement ratio	35%	45%
Community Rate	Users joining a community	60%	72%
Goal Completion	Goals completed vs. created	28%	40%
Buddy Verification Rate	Tasks verified on time	75%	85%
Store Purchases	Coin spends per active user/month	2.5	4.0
Retention (D30)	30-day retention rate	40%	52%

3. User Personas

Persona A — The Determined Solo

Profile: Priya, 29, Software Engineer. Introverted, self-motivated, has specific private goals (weight loss, language learning).

Needs: Full private mode, granular scheduling, progress analytics, calendar view.

Pain Points: Existing apps are too social or require sharing data. Wants rewards but values privacy.

Persona B — The Community Champion

Profile: Marcus, 34, Marketing Manager. Highly social, coaches a fitness group, thrives on shared wins.

Needs: Community creation, goal templates, leaderboards, buddy management, proof verification.

Pain Points: Current group chats lack structure. Needs accountability without administrative burden.

Persona C — The Goal Seeker

Profile: Ananya, 22, University Student. Not sure where to start with personal development.

Needs: Goal library, community onboarding, clear task cards, gamification to stay engaged.

Pain Points: Analysis paralysis, quitting when motivation dips. Needs social reinforcement.

Persona D — The Community Admin

Profile: Jordan, 41, Wellness Coach. Runs accountability programs professionally.

Needs: Admin controls, configurable XP rates, multiple community goals, buddy rotation control.

Pain Points: No tool combines accountability + gamification + professional control level.

4. Functional Requirements

4.1 Goal Management

FR-GM-01: Goal Creation

- A user can create a Goal with: Name, Description, Category, Target Date/Quarter, Privacy Level (Private / Community), Status (Active / Paused / Completed / Archived).
- Goals may have tags (user-defined or community-defined) for filtering.
- Goals can be created from scratch OR imported from the Open Goal Library OR adopted from a community's goal set.

FR-GM-02: Goal Structure

- Each Goal is a container for one or more Tasks.
- Goals display a progress ring calculated from completed task occurrences vs. total scheduled task occurrences within the goal's timeline.
- Goals support multiple milestones as optional sub-markers within the timeline.

FR-GM-03: Goal Library

- A browsable, searchable Open Goal Library of curated community-contributed goal templates.
- Library entries include: Goal name, description, suggested task list, difficulty rating, estimated duration, tags, and adoption count.
- Any user can fork a library goal to their own dashboard and customize it.
- Community admins can maintain a private community goal library visible to community members only.

4.2 Task Management

FR-TM-01: Task Configuration

- A Task belongs to exactly one Goal but can be pinned to multiple calendar views.
- Task attributes: Name, Description, Goal (parent), Type (One-time | Recurring), Tags, Priority, XP Value, Coin Value, Proof Required (Yes/No), Proof Type (Photo | Text | Link | Any).
- Recurring tasks define their recurrence rule: Daily, Weekly (select days), Monthly (select dates), Custom (cron-like input for power users).
- Each task instance can be scheduled to a specific time window (start time + optional duration).
- Tasks can be marked as Optional (still earns partial XP) or Mandatory (failure affects buddy XP).

FR-TM-02: Task States

Scheduled	Task is planned for a future date/time.
Pending	Task window is active and awaiting completion.
Completed (Unverified)	User marked done; awaiting buddy verification if required.
Completed (Verified)	Buddy confirmed completion. XP and coins distributed.
Skipped	User explicitly skipped. No XP. Buddy notified.
Missed	Window elapsed without action. Streak broken.
Failed (Rejected)	Buddy rejected proof. No XP. Appeal available.

FR-TM-03: Task Tags

- Tags are user-defined strings (e.g., 'morning', 'fitness', 'deep-work').
- Community-level tags are available to all community members and managed by admins.
- Tags are color-coded and filterable on dashboard, calendar, and leaderboard views.

4.3 Scheduling & Calendar View

FR-CAL-01: Calendar Interface

- A full Calendar screen (month / week / day views) displaying Task cards plotted by their scheduled date/time.
- Task cards on the calendar are color-coded by Goal, Priority, or Status (user-selectable preference).
- Drag-and-drop rescheduling: dragging a task card to a new date/time slot reschedules it. Recurring instances ask the user if they want to reschedule just this occurrence or all future occurrences.
- Time conflicts are highlighted in real-time during drag.
- Mini-map view: a condensed dot-per-task view when zoomed out to monthly.

FR-CAL-02: Scheduling Rules

- Goal timelines render as a Gantt-style bar across the calendar header row.
- The system warns when a task is scheduled outside its parent goal's timeline.
- Community-shared goals display community member aggregated task progress on the calendar (anonymized or attributed, per privacy settings).

4.4 Gamification System

FR-GAM-01: XP Economy

Task Completion (Solo)	Base XP defined on task config. Multiplied by streak bonus.
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Task Completion (Community, Verified)	Base XP x 1.0 (full). Streak bonus applies.
Buddy Verification (correct, on-time)	Configurable percentage of task XP (default: 30%). Admin can set 10-50%.
Buddy Rejection (false positive caught)	Small bonus XP to reporting user on appeal win.
Goal Completion Bonus	Flat XP bonus + coin bonus when all goal tasks are finished.
Streak Multipliers	3-day: 1.1x 7-day: 1.25x 14-day: 1.5x 30-day: 2.0x.
Community Rank Bonus	Monthly leaderboard rank 1-3 earns bonus coins (not XP).

FR-GAM-02: Level System

- XP accumulates globally per user across all goals.
- Level thresholds follow an exponential curve (L1: 0-500 XP, L2: 500-1500 XP, L3: 1500-3500 XP, etc.).
- Each level-up triggers an in-app celebration animation and coin reward.
- User's level is displayed on their profile, leaderboard entry, and buddy panel.

FR-GAM-03: Coins & Reward Store

- Coins are earned alongside XP (configurable coin-per-XP ratio, default 1:1).
- Coins are spent in the Reward Store for cosmetic items: Titles, Avatars, Profile Backgrounds, Task Card Themes, Goal Badge Overlays.
- Solo users earn 60% of normal coin rewards (configurable platform-level default).
- Reward Store items are categorized by rarity: Common, Rare, Epic, Legendary.
- Limited-edition seasonal items rotate quarterly.
- No paid currency. No pay-to-win. All store items are cosmetic only.

4.5 Community System

FR-COM-01: Community Creation & Structure

- Any user can create a Community with: Name, Description, Avatar, Privacy (Open / Invite-only / Private), Max Member Cap, Community Goals (one or more), Tags.
- Communities support multiple concurrent Goals. Members can participate in any or all community goals.
- Community Roles: Admin, Moderator, Member.

FR-COM-02: Buddy System

- When a user joins a community goal, they are assigned a buddy from the pool of active participants.

- Buddy assignment algorithm: round-robin by default, with smart pairing (similar level, similar timezone) as a premium option.
- Buddy rotation: admin configures the rotation interval (weekly, bi-weekly, monthly, or fixed).
- A user may have one buddy per community goal simultaneously.
- Buddy pairs are visible to each other on their dashboard in the Buddy Panel.

FR-COM-03: Proof & Verification

- When a task has Proof Required enabled, the completing user must upload proof at task completion (photo, text note, URL, or any type as configured).
- Proof submission triggers a push notification to the assigned buddy.
- Buddy has a configurable time window (default 24 hrs, configurable 6-72 hrs) to Approve or Reject with a comment.
- If buddy fails to verify in time, the system auto-approves and logs the missed verification (repeated misses reduce buddy's verification score).
- An appeal system allows users to escalate rejections to a community moderator.

FR-COM-04: Leaderboard

- Each community has a Leaderboard ranked by XP earned within that community's context (not global XP).
- Leaderboard time windows: All Time / This Month / This Week.
- Leaderboard displays: Rank, Username, Avatar, Level, Community XP, Goal Completion %, Streak.
- Top 3 at month-end receive a bonus coin award and an optional community badge.

4.6 Notifications & Reminders

- Push notifications (via Web Push API / FCM): task reminders, buddy requests, proof submission requests, verification expiry warnings, level-up celebrations.
- In-app notification center with read/unread states.
- User-configurable notification preferences per notification type.
- Daily digest email option (configurable frequency).

4.7 Solo Mode

- All goal and task management features available in solo mode.
- No buddy assignment, no proof verification, no community leaderboard.
- XP and coins earned at reduced rate (platform-configurable, default: 60% of community rates).
- Solo users can still access the Open Goal Library.
- Private goals are fully invisible to all other users, including community members.

5. UX & Design Requirements

5.1 Design Philosophy

- Mobile-first: designed for 375px-430px viewport as the canonical experience, gracefully scaling to tablet and desktop.
- Dark/Light mode support (system-default + manual toggle).
- Micro-animations on key interactions: task completion, level-up, coin earn, streak milestones.
- Accessibility: WCAG 2.1 AA compliant. Minimum contrast ratios, screen reader support, focus management.
- Game-inspired visual language without sacrificing clarity: subtle textures, XP bars, badge systems, particle effects on celebration moments.

5.2 Screen Inventory

Screen ID	Screen Name	Priority	Purpose
SCR-01	Onboarding / Welcome	P0	User understands value, signs up, sets first goal.
SCR-02	Dashboard / Home	P0	Today's tasks, active goals, buddy status at a glance.
SCR-03	Goal Detail View	P0	All tasks under a goal, progress ring, timeline.
SCR-04	Task Configuration Modal	P0	Create/edit a task with full scheduling options.
SCR-05	Calendar View	P0	All tasks across goals in a drag-and-drop calendar.
SCR-06	Community Hub	P0	Discover, join, or manage communities.
SCR-07	Community Detail	P0	Community goals, leaderboard, buddy panel, members.
SCR-08	Buddy Panel	P1	Buddy's progress, send encouragement, verify proofs.
SCR-09	Proof Submission	P0	Submit proof of task completion with media/text.
SCR-10	Proof Review	P0	Buddy reviews proof and approves/rejects.
SCR-11	Leaderboard	P1	Community-ranked XP standings with filters.
SCR-12	Reward Store	P1	Browse and purchase cosmetic items with coins.
SCR-13	User Profile	P1	Stats, badges, level, equipped cosmetics.
SCR-14	Goal Library	P2	Browse and adopt curated community goals.
SCR-15	Notifications Center	P1	Unified notification feed.
SCR-16	Settings	P1	Account, notifications, privacy, theme.

SCR-17	Analytics Dashboard	P2	Progress charts, completion trends, streak history.
SCR-18	Community Admin Panel	P1	Manage members, goals, buddy rules, XP config.

5.3 Dashboard (SCR-02) — Detailed Spec

- Top bar: XP bar (current level progress), coin count, notification bell.
- Today's Tasks strip: horizontally scrollable task cards sorted by time. Cards show: task name, parent goal color, time, status chip, proof indicator.
- Active Goals grid: compact cards with progress ring, goal name, next task, days remaining.
- Buddy Panel snapshot: buddy avatar, their today completion %, next verification due.
- Streak widget: current streak count, record streak, flame icon animation on active streak.
- Community Quick Links: first 3 communities with unread badge counts.

5.4 Calendar View (SCR-05) — Detailed Spec

- Default: weekly view with time-grid (30-min slots from 6:00 AM to midnight).
- Task cards on calendar: color = parent goal color. Height = proportional to task duration if set.
- Drag handle on task card. On drag-start, potential drop zones highlight.
- On drop: if one-time task, reschedule directly. If recurring: modal asks 'Move this occurrence only / Move all future / Move all'.
- Day column header shows: date, day of week, total task count, completion percentage dot indicator.
- Goal timeline bar rendered at the very top of the calendar spanning its start-to-end dates.
- Month view: mini task dots below date, clicking date expands a day popover.

5.5 Task Configuration (SCR-04) — Detailed Spec

- Form fields: Task Name, Description (rich text), Parent Goal (searchable select), Type toggle (One-Time / Recurring).
- Recurring options: Frequency dropdown, Day picker (weekly), Date picker (monthly), Custom cron (advanced toggle).
- Time scheduling: Start Time picker, Duration (optional), Reminder offset.
- Tags: multi-select chip input with autocomplete from user/community tag pool.
- Proof settings: toggle Proof Required, select Proof Type.
- XP value: editable with min/max guard (community admins can lock this).
- Mandatory toggle: if on, skip/miss affects buddy's standing.
- Preview: live calendar preview showing when recurrences will fall.

6. Technical Architecture

6.1 Technology Stack

Layer	Technology	Rationale	Version Target
Frontend	React + Vite (PWA)	Component ecosystem, PWA support, fast HMR	React 19+
State Management	Zustand + React Query	Lightweight global state + server state caching	Latest stable
Styling	Tailwind CSS + shadcn/ui	Utility-first, consistent design system	Tailwind v4
Mobile PWA	Workbox + Web App Manifest	Offline support, install prompts, push notifications	Latest
Backend	Go (Golang)	High concurrency, low latency, cloud-native	Go 1.22+
API Layer	REST + GraphQL (hybrid)	REST for CRUD, GraphQL for complex queries	gqlgen
Auth	Supabase Auth / Clerk	OAuth2, JWT, social login, MFA support	Latest
Primary DB	PostgreSQL (via Supabase)	ACID, relational, RLS for multi-tenancy	PG 16
Cache	Redis (Upstash serverless)	Session cache, leaderboard sorted sets, rate limiting	Redis 7+
File Storage	Supabase Storage / S3	Proof media uploads, avatars, backgrounds	Cloudflare R2
Push Notifications	Firebase Cloud Messaging	Web push, background sync	FCM v1 API
Search	PostgreSQL FTS / Typesense	Goal library search, community discovery	Typesense 0.25+
Email	Resend / SendGrid	Transactional email, digests	Latest
Infra (MVP)	Railway / Render	Simple Go deployment, managed Postgres	Managed
Infra (Scale)	GCP / AWS ECS + RDS	Autoscaling, global CDN, managed services	Kubernetes
CI/CD	GitHub Actions	Automated test, lint, deploy pipelines	Latest
Monitoring	OpenTelemetry + Grafana	Distributed tracing, metrics, alerting	OTel 1.x

6.2 System Architecture Overview

GoalKeeper follows a layered, modular architecture optimized for horizontal scalability. The frontend is a React PWA served via a CDN edge network. The backend is a stateless Go service fleet behind a load balancer, communicating with a PostgreSQL primary (with read replicas) and a Redis cluster. All media is stored in object storage with CDN delivery.

Architecture Layers (Top to Bottom)

- 1. Client Layer: React PWA (CDN-delivered, installable, offline-capable via Service Worker)
- 2. API Gateway: Nginx / Cloudflare Workers (rate limiting, auth validation, routing, DDoS protection)
- 3. Application Layer: Go microservices (Goals, Tasks, Users, Community, Gamification, Notifications, Store, Analytics)
- 4. Data Layer: PostgreSQL (primary + read replicas) | Redis (cache + leaderboard) | Object Storage (proof media)
- 5. Event Bus: NATS / Cloud Pub/Sub for async notifications, XP distribution, buddy alerts
- 6. AI Module (Future): Standalone inference service (Python/FastAPI) consuming data via internal gRPC API

6.3 Go Service Modules

users-service	Registration, auth integration, profile, settings, cosmetic inventory management.
goals-service	CRUD for goals, library management, progress calculation, milestone tracking.
tasks-service	Task CRUD, recurrence engine, scheduling rules, state machine management.
community-service	Community CRUD, membership, roles, buddy assignment algorithm, rotation scheduler.
gamification-service	XP engine, coin engine, level calculation, streak tracking, reward distribution.
verification-service	Proof submission, buddy notification, verification workflow, appeal handling.
store-service	Item catalog, inventory management, purchase transactions, coin deduction.
notification-service	Push notification dispatch, email, in-app notification feed, digest scheduling.
search-service	Goal library indexing, community discovery, user search (Typesense integration).
analytics-service	Progress aggregation, trend computation, heatmap data preparation for AI feed.

6.4 Key Database Schema

users

id UUID PK | email TEXT UNIQUE | display_name TEXT | avatar_url TEXT | level INT DEFAULT 1 | total_xp BIGINT DEFAULT 0 | coins BIGINT DEFAULT 0 | streak_current INT | streak_best INT | solo_mode BOOL | created_at TIMESTAMPTZ | updated_at TIMESTAMPTZ

goals

id UUID PK | user_id UUID FK(users) | community_id UUID FK(communities) NULLABLE | title TEXT | description TEXT | category TEXT | privacy ENUM(private,community,public) | status ENUM(active,pause,completed,archived) | target_date DATE | library_ref UUID NULLABLE | created_at | updated_at

tasks

id UUID PK | goal_id UUID FK(goals) | user_id UUID FK(users) | title TEXT | type ENUM(one_time,recurring) | recurrence_rule JSONB | scheduled_at TIMESTAMPTZ | duration_minutes INT | status ENUM(scheduled,pending,completed_unverified,completed_verified,skipped,missed,failed) | xp_value INT | coin_value INT | is_mandatory BOOL | proof_required BOOL | proof_type ENUM(photo,text,link,any) | tags TEXT[] | priority ENUM(low,medium,high) | created_at | updated_at

communities

id UUID PK | name TEXT | description TEXT | avatar_url TEXT | privacy ENUM(open,invite_only,private) | max_members INT | buddy_rotation_days INT | buddy_xp_ratio DECIMAL(4,2) DEFAULT 0.30 | coin_solo_ratio DECIMAL(4,2) DEFAULT 0.60 | created_by UUID FK(users) | created_at | updated_at

buddy_assignments

id UUID PK | community_id UUID FK(communities) | goal_id UUID FK(goals) | user_id UUID FK(users) | buddy_id UUID FK(users) | rotation_start TIMESTAMPTZ | rotation_end TIMESTAMPTZ | verification_score DECIMAL(5,2) DEFAULT 100.0 | created_at

proof_submissions

id UUID PK | task_id UUID FK(tasks) | submitter_id UUID FK(users) | buddy_id UUID FK(users) | media_url TEXT[] | note TEXT | status ENUM(pending,approved,rejected,auto_approved,appealed) | buddy_comment TEXT | submitted_at TIMESTAMPTZ | reviewed_at TIMESTAMPTZ | expires_at TIMESTAMPTZ

xp_ledger (partitioned by created_at monthly)

id UUID PK | user_id UUID FK(users) | amount INT | type
ENUM(task_completion,buddy_verification,level_bonus,goal_completion,streak_bonus,community_rank)
| ref_id UUID | community_id UUID NULLABLE | created_at TIMESTAMPTZ

store_items

id UUID PK | name TEXT | description TEXT | category
ENUM(title,avatar,background,card_theme,goal_badge) | rarity
ENUM(common,rare,epic,legendary) | coin_cost INT | asset_url TEXT | is_limited BOOL |
available_from TIMESTAMPTZ | available_until TIMESTAMPTZ

7. API Design

7.1 REST API Conventions

Base URL	https://api.goalkeeper.app/v1
Auth	Bearer JWT (Authorization header). RS256 signed.
Content-Type	application/json
Pagination	Cursor-based (?cursor=<token>&limit=<n>)
Errors	RFC 7807 Problem Details: { code, message, details }
Rate Limiting	100 req/min per user (Redis token bucket)
Versioning	URL path versioning /v1, /v2

7.2 Key API Endpoints

Method	Endpoint	Description	Auth Required
POST	/auth/register	User registration	No
POST	/auth/login	Login, returns JWT + refresh token	No
GET	/users/me	Get current user profile	Yes
PATCH	/users/me	Update profile / settings	Yes
GET	/goals	List user goals (filters: status, tag, date)	Yes
POST	/goals	Create a new goal	Yes
GET	/goals/:id	Get goal detail + associated tasks	Yes
PATCH	/goals/:id	Update goal attributes	Yes
DELETE	/goals/:id	Archive or delete goal	Yes
GET	/goals/library	Browse open goal library (paginated)	Optional
POST	/goals/library/:id/adopt	Fork library goal to user dashboard	Yes
GET	/tasks	List tasks (filter by goal, date, status, tag)	Yes
POST	/tasks	Create task under a goal	Yes
PATCH	/tasks/:id	Update task configuration	Yes
POST	/tasks/:id/complete	Mark task complete + attach proof	Yes
PATCH	/tasks/:id/reschedule	Reschedule task occurrence(s)	Yes
GET	/communities	Discover / list communities	Optional
POST	/communities	Create a community	Yes

POST	/communities/:id/join	Join a community	Yes
GET	/communities/:id/leaderboard	Get community leaderboard	Yes
GET	/communities/:id/buddy	Get current buddy assignment	Yes
GET	/proofs/pending	List proof submissions awaiting review	Yes
POST	/proofs/:id/review	Approve or reject proof with comment	Yes
GET	/gamification/me	XP, level, coins, streak summary	Yes
GET	/store/items	List store items (filter by category, rarity)	Yes
POST	/store/purchase/:itemId	Purchase store item with coins	Yes
GET	/notifications	List user notifications (paginated, cursor)	Yes
PATCH	/notifications/:id/read	Mark notification as read	Yes
GET	/calendar	Task instances within a date range (?from=&to=)	Yes

8. User Stories & Acceptance Criteria

8.1 Story Epics

Epic ID	Epic Name	Priority	Story Range
EP-01	Authentication & Onboarding	P0	US-01 to US-05
EP-02	Goal Lifecycle Management	P0	US-06 to US-14
EP-03	Task Scheduling & Calendar	P0	US-15 to US-24
EP-04	Community & Buddy System	P0	US-25 to US-38
EP-05	Proof & Verification	P0	US-39 to US-46
EP-06	Gamification & Rewards	P1	US-47 to US-57
EP-07	Notifications & Reminders	P1	US-58 to US-63
EP-08	Goal Library	P2	US-64 to US-69
EP-09	Analytics & Progress	P2	US-70 to US-76
EP-10	Admin & Community Management	P1	US-77 to US-85

8.2 Representative User Stories

US-06 | Epic EP-02: Goal Lifecycle | Priority: P0

Story: As a user, I want to create a Goal with a name, description, end date, and privacy level so that I can define what I am working toward.

Acceptance Criteria:

- Form validates name (required, max 120 chars), description (optional, max 1000 chars), target date (must be future), privacy (required).
- On submit, goal appears in user's dashboard within 1 second.
- Goal progress ring displays at 0% on creation.
- Private goal is not visible to any other user or community member.

US-28 | Epic EP-04: Community & Buddy | Priority: P0

Story: As a community member, I want to be assigned a buddy automatically when I join a community goal so that I have an accountability partner.

Acceptance Criteria:

- On joining a community goal, system assigns a buddy within 5 seconds.
- Both user and buddy receive a push notification with each other's profile card.
- Buddy Panel on dashboard shows buddy's name, avatar, level, today's task status.
- If no eligible buddy is available, user is placed in a queue and notified when assigned.

US-47 | Epic EP-06: Gamification | Priority: P1

Story: As a user, I want to earn XP and coins when I complete a verified task so that I feel rewarded for my progress.

Acceptance Criteria:

1. On verification approval, XP and coins are credited within 3 seconds.
2. XP bar on dashboard visually animates to the new value.
3. Buddy simultaneously receives their configured percentage of XP with a notification.
4. Solo task earns 60% of base XP/coins (configurable platform default).
5. Streak multiplier is applied correctly before final XP is awarded and shown in the reward toast.

US-39 | Epic EP-05: Proof & Verification | Priority: P0

Story: As a buddy, I want to receive a notification and review my partner's proof submission so that I can verify their task completion accurately.

Acceptance Criteria:

1. Buddy receives push notification within 30 seconds of proof submission.
2. Proof review screen shows media/text, task name, submission time, and remaining review window.
3. Buddy can Approve (XP distributed) or Reject (with mandatory comment, no XP, appeal option shown to submitter).
4. If no action taken within the configured window, proof auto-approves and buddy's verification score decrements.

9. Non-Functional Requirements

9.1 Performance

NFR-P01	API p95 response time < 300ms for all CRUD endpoints under normal load.
NFR-P02	Leaderboard reads < 100ms (Redis sorted set / ZSET).
NFR-P03	Calendar query for 30-day range returns within 500ms for up to 500 task instances.
NFR-P04	Frontend Lighthouse Performance score > 85 on mobile.
NFR-P05	PWA initial load (First Contentful Paint) < 2.5s on 4G mobile.
NFR-P06	Media proof uploads: client-side compression before upload, max 5MB post-compression.

9.2 Scalability

NFR-S01	Backend must support 100,000 concurrent users with horizontal pod scaling (Kubernetes HPA).
NFR-S02	Database design must support 10M+ users without schema changes (UUID PKs, partitioned xp_ledger and task_instances tables).
NFR-S03	Leaderboard service scales to 10,000 community members using Redis ZSET natively.
NFR-S04	Notification service decoupled via event bus to prevent cascading failures on core task operations.

9.3 Security

NFR-SEC01	All API endpoints authenticated via JWT (RS256). Tokens expire in 1 hour; refresh tokens valid for 30 days.
NFR-SEC02	Row-level security (RLS) enforced at Postgres level. Users cannot access other users' private data.
NFR-SEC03	Media upload URLs are pre-signed with 5-minute expiry. Stored media accessed via signed CDN URLs only.
NFR-SEC04	OWASP Top 10 compliance. SAST scans in CI pipeline (Gosec for Go, ESLint Security for React).
NFR-SEC05	Rate limiting: 100 req/min per authenticated user; 20 req/min for anonymous endpoints.
NFR-SEC06	GDPR compliance: right to erasure, data export endpoint, consent management for analytics.
NFR-SEC07	All data encrypted in transit (TLS 1.3) and at rest (AES-256).

9.4 Reliability & Availability

NFR-R01	Target SLA: 99.5% uptime (MVP) scaling to 99.9% post-Series A infrastructure upgrade.
NFR-R02	Graceful degradation: if notification service is down, core task completion continues unaffected.
NFR-R03	Buddy verification auto-approval fallback fires if verification service is unreachable for > 1 hour.
NFR-R04	Database failover: Postgres read replica promotes to primary within 60 seconds.
NFR-R05	PWA offline mode: task completion queued locally (IndexedDB) and synced on reconnect.

10. Phased Delivery Roadmap

Phase 0 — Foundation (Weeks 1-4)

Infrastructure, Auth, Core Data Models

Goals: Set up monorepo, CI/CD pipeline, Go service scaffolding, PostgreSQL schema migrations, React PWA shell, Auth integration, user profile service.

Deliverables: Deployable Go API skeleton, React app with working auth flow, database migrations, dev environment documentation.

Definition of Done: A new developer can clone, install, and run the full stack locally in < 30 minutes.

Phase 1 — Core MVP (Weeks 5-12)

Goals, Tasks, Calendar, Solo Mode

Goals: Full goal CRUD, task CRUD with recurrence engine, calendar view with drag-and-drop rescheduling, solo gamification (XP/coins/levels/streaks), dashboard, push notification reminders.

Deliverables: Functional app for solo users. Internal alpha. First user testing round with 20 participants.

Success Gate: DAU/MAU ratio > 25% among alpha testers. Task completion rate > 40% among alpha cohort.

Phase 2 — Community & Accountability (Weeks 13-20)

Community Hub, Buddy System, Proof & Verification, Leaderboards

Goals: Community creation and joining flows, buddy assignment algorithm (round-robin + smart), proof submission and verification workflow, leaderboard (Redis ZSET), community admin panel, configurable XP/coin ratios.

Deliverables: Closed beta with 5-10 real communities. Load testing to 1,000 CCU. Bug bash week.

Success Gate: Buddy verification rate > 70%. No critical P0 bugs in core accountability flow.

Phase 3 — Reward Economy & Store (Weeks 21-26)

Reward Store, Cosmetics, Goal Library

Goals: Reward store with item catalog, coin transactions, inventory system, first cosmetic item set (10 avatars, 5 backgrounds, 20 titles, 3 card themes), Open Goal Library seeded with 50 curated goals, tag system.

Deliverables: Public beta launch. PWA installability live on iOS and Android. Marketing site live.

Success Gate: 1,000 registered users in first 2 weeks. Store purchase rate > 20% of active users.

Phase 4 — Analytics & AI Foundation (Weeks 27-36)

Analytics Dashboard, AI Integration Points

Goals: User analytics dashboard (progress charts, completion heatmaps, streak history, goal trajectory), data pipeline to analytics-service, AI module API contract defined, initial AI feature: smart task scheduling suggestion based on historical patterns.

Deliverables: v1.0 GA release. Investor demo build. AI module proof-of-concept with backtesting report.

Phase 5 — Scale & Enterprise (Post-Launch)

Enterprise Tier, API Monetization, Advanced AI

Goals: White-label community tier for enterprises, enterprise SSO (SAML), public API key access for third-party integrations, advanced AI goal coaching (personalized task suggestions, risk-of-quit prediction, adaptive difficulty), global cross-community leaderboards.

11. AI Module — Future Scope

11.1 Architecture Approach

The AI module is designed as a standalone microservice (Python / FastAPI preferred for ML ecosystem compatibility) that consumes data from the analytics-service via an internal gRPC or REST API. This separation ensures AI model updates do not require frontend or core backend deploys, and allows the inference service to scale independently.

11.2 Planned AI Capabilities

Feature	Description	Data Required	Timeline
Smart Scheduling	Suggest optimal task times based on historical completion patterns.	Task timestamps, completion rate by time-of-day	Phase 4
Goal Risk Prediction	Alert user when goal trajectory suggests missing the target date.	Task completion velocity, goal timeline	Phase 4
Adaptive XP Tuning	Recommend XP value adjustments based on difficulty signals.	Completion rate, time taken, skip rate per task	Phase 5
AI Buddy Matching	Intelligent buddy pairing based on complementary behavioral profiles.	User behavioral graph, completion history	Phase 5
Natural Language Goal Input	Users describe goals in plain language; AI structures into Goal + Tasks.	LLM inference (Claude API / GPT-4o)	Phase 5
Progress Coaching	Personalized weekly insight messages based on behavioral patterns.	Full user event history, goal progress	Phase 5

11.3 Data Privacy for AI

- All AI training uses anonymized, aggregated data only. No personal identifiers in model training sets.
- User opt-out available for behavioral data collection used for AI personalization features.
- On-device inference explored for scheduling suggestions (TensorFlow.js) to minimize data transmission.
- AI feature set clearly labeled in UI. Users can see what data is used for each AI feature.

12. Risks & Mitigations

Risk	Impact	Likelihood	Mitigation Strategy
Buddy ghosting / non-verification	High — breaks core accountability loop	High	Auto-approve fallback, verification score penalties, buddy swap option, escalation to moderator.
XP gaming / fake proof submissions	High — economy integrity	Medium	Community moderation tools, report system, machine-based proof scan in Phase 4.
Low community adoption at launch	High — community features worthless without users	Medium	Strong solo mode, seeded communities by team, influencer / community manager partner program.
Backend performance at scale	High — user experience degradation	Medium	Load testing Phase 2, Redis caching, read replicas, autoscaling configuration.
PWA push notification limitations (iOS)	Medium — engagement gap on iOS users	High	Monitor Apple PWA Push API, fallback to email and in-app notification for iOS users.
Goal Library content quality	Medium — poor templates damage trust	Medium	Editorial curation team, community upvote/report system, quality scoring algorithm.
Scope creep delaying MVP	High — missed market window	Medium	Strict phase-gating, MoSCoW prioritization each sprint, weekly scope review with PM.
GDPR compliance gaps	High — regulatory risk in EU	Low	Legal review before Phase 3 public launch, Data Protection Impact Assessment (DPIA).

13. Open Questions & Decisions Required

OQ-01	Monetization model: Freemium (what is behind paywall?) vs. pure free with enterprise tier? Decision required before Phase 3.
OQ-02	Buddy XP default ratio: 30% community default — should admin cap range be 10%-50% or wider? Requires economy review.
OQ-03	PWA vs. native app: Should we invest in React Native for Phase 5 or remain PWA-only?
OQ-04	Proof storage retention: How long are proof media files retained? 90 days? 1 year? Directly impacts storage cost projections.
OQ-05	Community size limits: Is 500 members per community the right cap? Affects buddy pool quality and leaderboard competitiveness.

OQ-06	Open Goal Library governance: Who curates? Community vote vs. editorial team vs. AI scoring?
OQ-07	Seasonal cosmetics pricing: What coin costs prevent inflation while keeping items desirable?
OQ-08	GDPR data residency: Do we need EU-region infrastructure at launch for EU users?

14. Glossary

Goal	A long-term objective with a defined timeline, composed of one or more Tasks.
Task	A specific, schedulable unit of work that contributes to completing a Goal.
Buddy	A community-assigned accountability partner who verifies task completion proof.
XP (Experience Points)	Points earned for completing tasks and verifying buddy tasks. Drive level progression.
Coins	In-app currency earned alongside XP, spent in the Reward Store for cosmetics.
Proof	Evidence (photo, text, link) submitted by a user to confirm task completion.
Verification	The act of a buddy reviewing and approving or rejecting submitted proof.
Streak	Consecutive days with at least one verified task completion. Drives XP multipliers.
Open Goal Library	A public repository of curated, community-contributed Goal + Task templates.
Round Robin	A buddy rotation method that cycles through community members in sequence.
PWA	Progressive Web Application — a web app installable on mobile/desktop with native-like features.
RLS	Row-Level Security — PostgreSQL feature restricting data access at the database level.
HPA	Horizontal Pod Autoscaler — Kubernetes mechanism for auto-scaling service replicas.
ZSET	Redis Sorted Set — data structure used for real-time leaderboard ranking by score.
Recurrence Rule	A JSON-encoded ruleset defining when a recurring task repeats (days, frequency, cron).

Verification Score	A per-buddy metric tracking timeliness and accuracy of their verification actions.
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